

NORVENICH, Germany

WMO: 105020

Lat: 50.8294N

Lon: 6.6603E

Elev: 439

StdP: 14.46

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.0	22.9	12.9	10.8	22.4	16.9	13.1	25.2	28.6	47.7	25.6	45.6	5.3	120	0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	87.9	67.4	83.9	66.3	80.2	64.8	69.5	82.9	67.8	80.1	66.3	77.3	7.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.1	94.3	73.9	63.6	89.6	72.2	62.2	85.1	71.0	33.7	83.0	32.4	80.2	31.2	77.1	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.2	18.1	15.5	DB	12.5	95.4	6.8	3.0	7.6	97.5	3.6	99.3	-0.2	101.0	-5.2	103.2
			WB	12.0	73.1	6.7	1.8	7.2	74.4	3.3	75.4	-0.5	76.4	-5.4	77.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	37.6	39.1	44.3	50.4	57.4	62.8	66.7	65.8	59.6	52.5	44.6	39.0	(6)
	DBStd	12.22	8.04	7.52	6.41	6.74	6.49	5.91	5.93	5.51	5.44	6.45	6.56	7.75	(7)
	HDD50	1560	388	308	195	77	11	0	0	0	1	49	184	347	(8)
	HDD65	5101	850	724	641	438	247	111	50	55	179	389	611	806	(9)
	CDD50	2192	3	4	19	90	242	385	517	490	289	125	22	6	(10)
	CDD65	261	0	0	0	1	12	46	103	80	18	1	0	0	(11)
	CDH74	2590	0	0	0	33	156	462	990	781	160	8	0	0	(12)
	CDH80	825	0	0	0	2	28	130	346	283	36	0	0	0	(13)
(14) Wind	WSAvg	6.7	8.0	7.8	7.6	6.6	6.2	6.1	6.3	5.9	6.0	6.3	6.7	7.4	(14)
(15) Precipitation	PrecAvg	27.7	2.0	2.0	1.9	1.7	2.2	2.8	2.8	2.9	2.1	2.2	2.3	2.4	(15)
	PrecMax	33.4	4.7	4.8	3.4	3.3	4.1	5.1	7.5	6.5	5.2	5.6	3.9	5.6	(16)
	PrecMin	22.2	0.3	0.3	0.2	0.0	0.9	0.6	1.3	1.1	0.4	1.1	0.0	0.6	(17)
	PrecStd	3.3	1.0	1.0	0.9	0.8	0.9	1.1	1.5	1.5	1.3	1.0	0.9	1.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.2	60.2	68.1	78.6	84.1	89.9	93.4	93.4	85.3	74.9	63.4	57.6	(19)
		MCWB	51.6	49.8	53.4	58.3	65.1	69.8	68.2	69.0	66.2	62.7	56.4	52.1	(20)
	2%	DB	53.8	55.2	62.6	72.9	78.1	83.9	88.0	87.5	78.6	69.5	59.2	55.2	(21)
		MCWB	49.0	48.2	50.9	56.3	61.7	66.7	67.5	67.3	63.8	60.0	53.6	50.6	(22)
	5%	DB	51.0	52.3	58.3	67.8	74.4	79.2	83.6	82.4	73.7	65.3	56.8	52.7	(23)
		MCWB	46.9	46.9	49.8	54.8	60.5	64.5	66.2	65.8	62.0	58.1	52.0	48.5	(24)
	10%	DB	48.3	49.9	54.8	62.9	70.2	75.2	79.5	77.5	69.7	62.3	53.9	50.0	(25)
		MCWB	44.5	45.5	48.1	52.8	58.7	63.0	64.8	64.2	60.3	56.3	49.7	46.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	51.3	54.9	60.7	67.5	71.7	72.2	71.0	67.6	63.4	57.2	53.2	(27)
		MCDB	56.2	56.7	64.1	73.1	79.3	85.3	86.6	86.5	79.5	71.3	62.1	56.4	(28)
	2%	WB	49.5	49.4	52.4	58.0	64.4	68.5	69.5	68.9	65.3	60.8	54.1	51.0	(29)
		MCDB	53.3	54.0	59.7	69.3	75.1	80.4	82.5	82.8	75.4	67.9	58.4	54.8	(30)
	5%	WB	47.2	47.6	50.5	55.6	61.8	66.3	67.9	67.0	63.2	58.9	52.2	48.8	(31)
		MCDB	51.0	52.0	56.8	65.3	71.1	76.8	80.3	79.3	71.8	64.4	56.2	52.2	(32)
	10%	WB	44.7	45.5	48.8	53.6	59.5	64.1	66.2	65.3	61.2	57.0	50.1	46.5	(33)
		MCDB	47.9	49.6	54.4	62.3	68.6	72.8	76.8	75.7	68.3	61.6	53.6	49.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	11.2	14.6	18.1	18.1	18.5	18.5	18.6	16.7	14.1	10.5	9.1	(35)
		MCDBR	11.8	14.8	22.1	26.7	25.6	26.3	26.5	27.9	23.8	18.8	12.9	11.0	(36)
	5% WB	MCWBR	9.4	10.2	12.9	13.4	12.7	12.7	10.5	11.5	11.6	11.1	9.2	8.5	(37)
		MCWBR	11.5	13.4	18.1	23.0	22.2	22.6	23.1	23.2	20.7	17.5	12.2	10.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.315	0.339	0.370	0.407	0.412	0.415	0.419	0.412	0.382	0.357	0.333	0.312	(40)
		taud	2.432	2.385	2.312	2.238	2.258	2.268	2.255	2.302	2.367	2.436	2.431	2.409	(41)
	Ebn at Noon		226	250	261	264	268	267	264	260	255	240	216	207	(42)
		Edh at Noon	19	26	33	40	41	41	41	37	31	24	19	17	(43)
(44) All-Sky Solar Radiation	RadAvg		237	455	812	1299	1541	1685	1596	1368	1009	590	275	181	(44)
	RadStd		37	87	124	151	143	157	193	132	124	58	36	25	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page